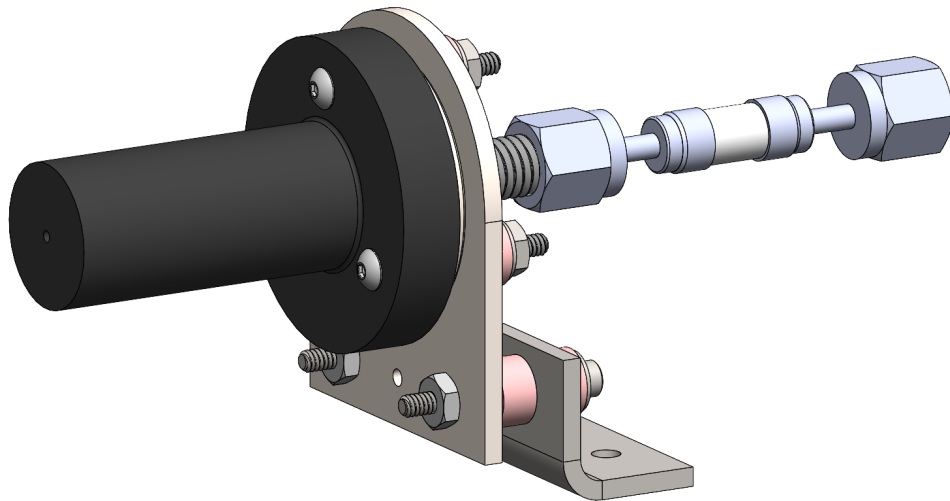


201-0140 Cathode Assembly

- **Manual**

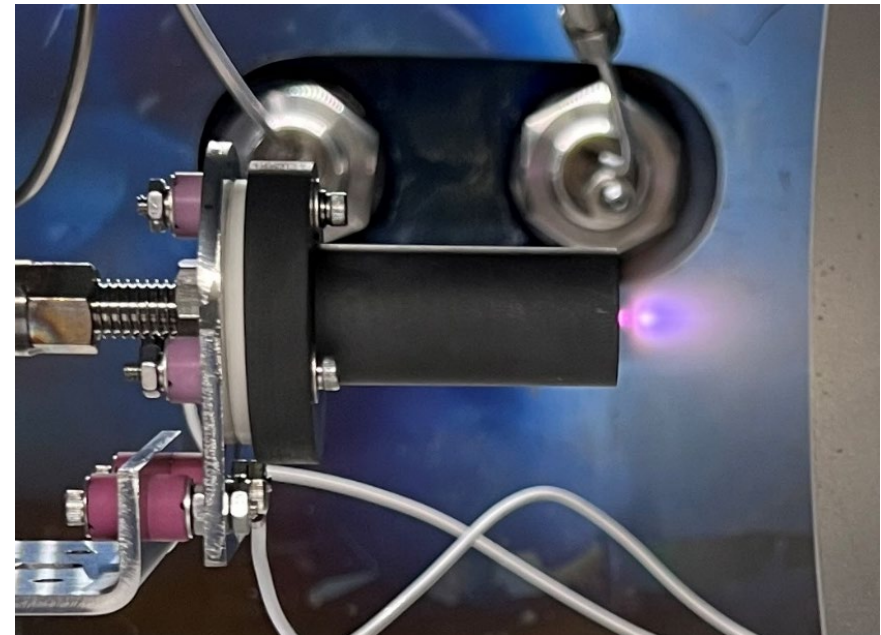
- Rev C : Nov. 15, 2024



201-0140 Cathode Assembly



info@plasmacontrols.com



2.0 A discharge current, 7.5 sccm argon

Setup

- This guide and hollow cathode assembly are indented for users already familiar with plasma source fundamentals and operation.
- Vacuum Environment
 - The cathode assembly should be operated in a high vacuum environment, with typical vacuum chamber base pressures in the 10^{-7} to 10^{-4} torr range. Vacuum pumping speeds may vary, with typical operating pressures in the 10^{-6} to 10^{-3} torr range for inert gasses like argon, krypton, and xenon.
- Gas Supply
 - Ensure a clean, leak free gas delivery flow system to the cathode. To facilitate long life operation, oxygen concentrations should be low, preferably in the < 100 ppm range.
- Electrical
 - The hollow cathode assembly is operated with user-supplied DC power supplies that can potentially be at **hazardous voltages**. Only technically competent personnel generally familiar with the principles of electrical safety should carry out operation of the hollow cathode assembly.

Application Example

HCA125 Cathode Assembly

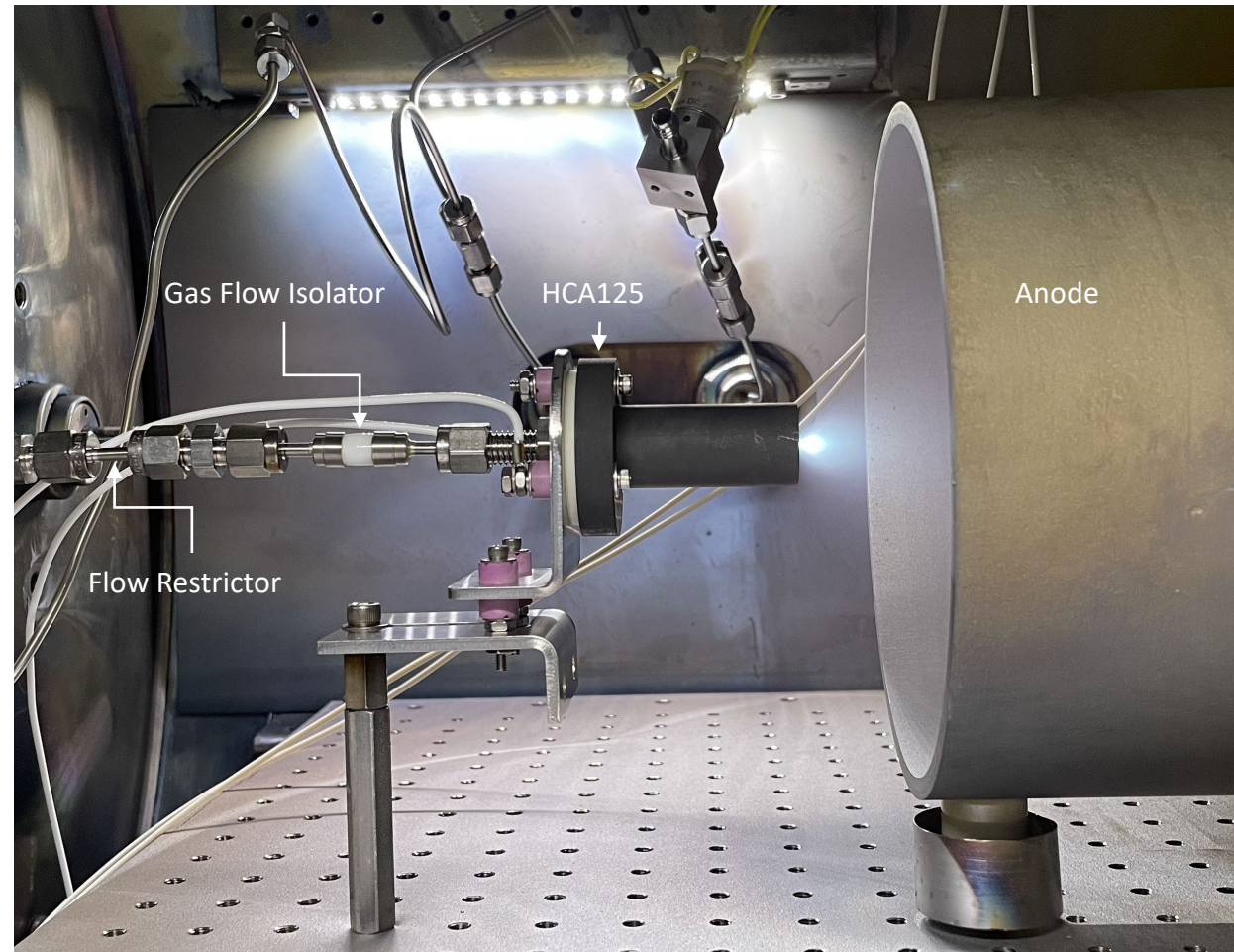
- PN 201-0140

Hollow Cathode Tip

- HCT125 1/8" OD

Flow Restrictor

- K2-7-SS 0.0071"
- Optional to assist ignition where a high flow rate MFC (100+ sccm) is not installed.



3.0 A discharge current, 1.0 sccm xenon

Typical Configuration

Keeper

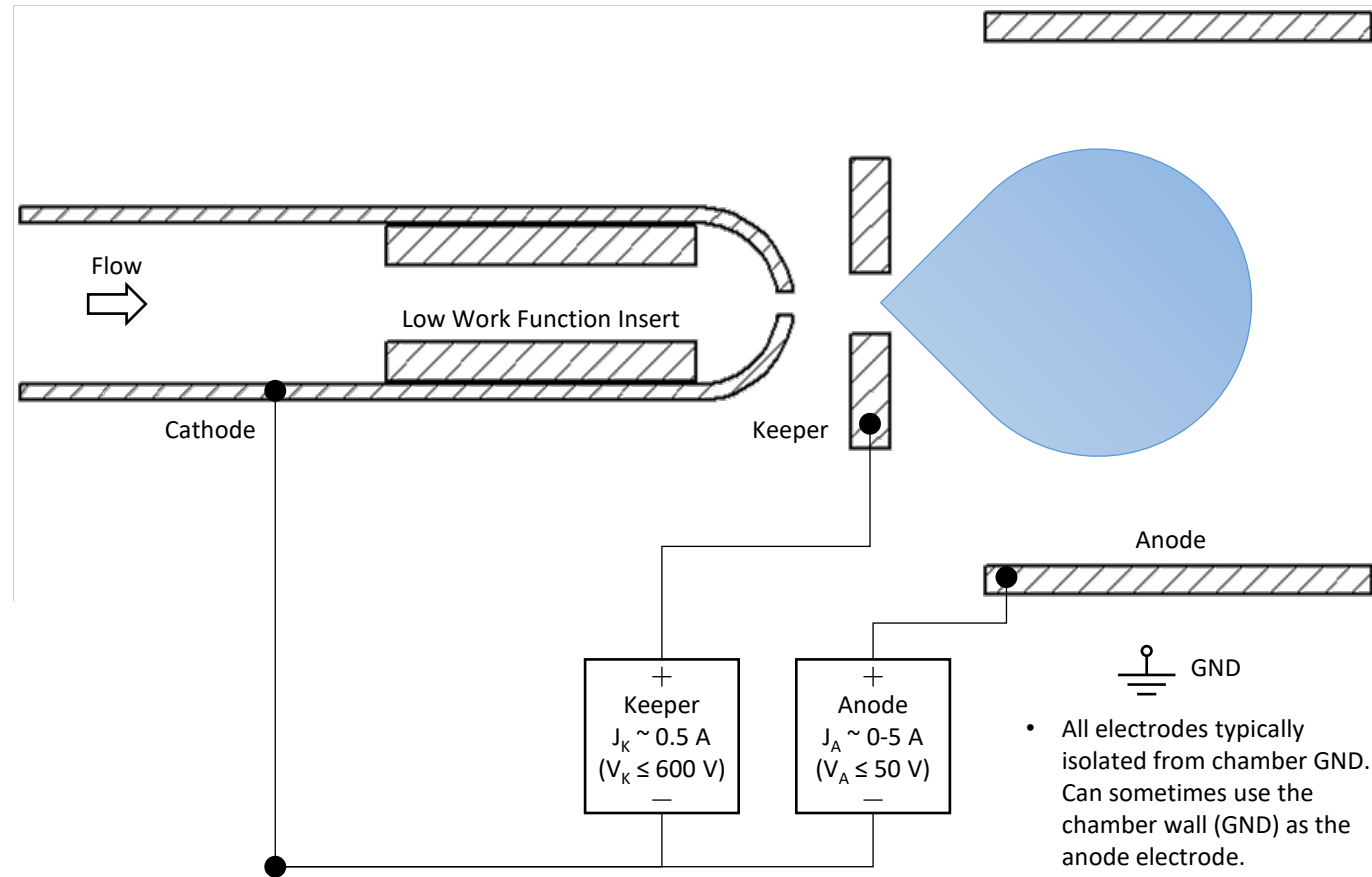
- Usually operated in current control mode.
- Typically apply output at $V_K \sim 600$ V & $J_K \sim 1.5$ A setpoints for starting the heaterless cathode.
- $J_K \leq 1.0$ A typ. while in operation.
 - $V_K \sim 5-15$ V

Anode

- $J_A = 0-5$ A for a 1/8" OD cathode.
- Applying $V_A = 100+$ V before/alongside V_K may assist cathode ignition.

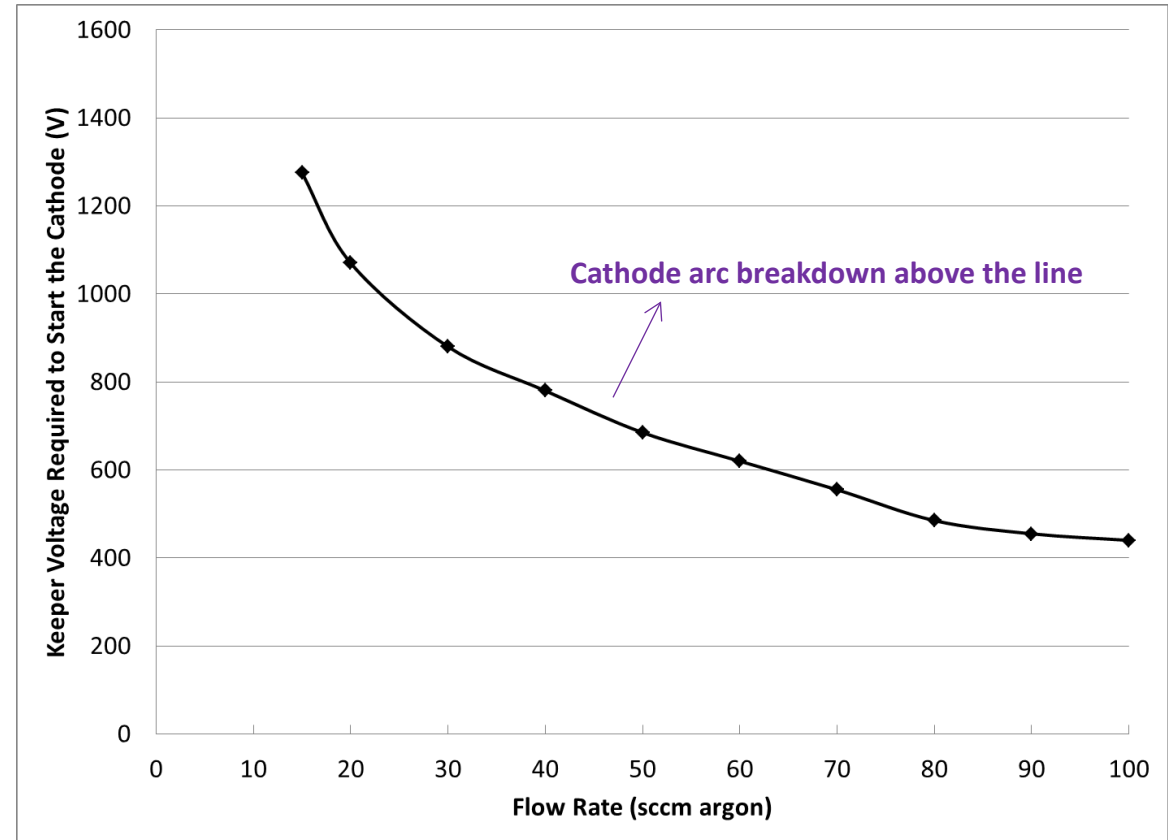
Insert

- Low work function, W-BaO
- Could be tubular or plug shaped based on tube diameter.



Ignition Schemes

- Starting the heaterless cathode requires a combination of elevated **flow rate** and **keeper voltage**.
- Method A**
 - Apply a constant keeper voltage (~600 V), and temporarily increase the gas flow (~100 sccm) to the cathode to initiate a breakdown.
 - Fixed Volume Release, **“FVR” Method** (preferred)
 - Useful if a high flow rate MFC or proportional valve is not available.
 - Set a fixed flow rate (e.g. 10 sccm)
 - Close a valve between the MFC and cathode, build up a bolus of gas for an amount of time (e.g. 15-30 seconds).
 - Manual, solenoid, pneumatic either on air or vacuum side.
 - Apply keeper voltage, 600 V & >1.0 A if available.
 - Open the valve and let the gas surge increase pressure in the cathode-keeper gap and initiate a breakdown for several seconds (5-10 seconds ideally) to rapidly heat the cathode.
- Method B**
 - Apply an elevated steady-state gas flow through the cathode (15 to 100 sccm in the plot), and then apply and/or increase the keeper voltage until the keeper discharge starts (from 425 V at high flow to 1300 V at low flow shown in this plot).
 - Sometimes, this method is more likely to result in **unwanted arcing** to occur in locations other than the cathode tip, i.e. near the upstream base of the assembly, which can **coat the (alumina) isolators** and cause shorting and unwanted preferential arcing.



Ignition, Operation, & Troubleshooting

- **Fixed Volume Release (FVR) Ignition**

- Set the MFC flow rate [e.g. 10 sccm, where 5-20 sccm is typ.].
- Close the shut off valve between the MFC and cathode.
 - This could be an external manual valve or a solenoid or pneumatic valve either on the air or vacuum side for example.
- Wait for the flow line / volume to fill with gas [30-120 seconds typ.]
 - The gas charge is known (flow rate x time).
 - In some cases, the flow lines may completely fill, indicated by the MFC flow rate tapering off as the pressure across the MFC equalizes.
- Apply the keeper supply output [e.g. 600 V and 1.5 A, where 600 V and ≥ 0.75 is typ.].
 - Higher setpoint, e.g. 2.0 A, usually helps ignition.
 - Note: If an anode power supply is present, or if there is only an anode supply, enabling that supply's output may also assist smooth ignition (e.g. 100V and 1-5 A).
- Open the shut off valve.
 - The pressure between the cathode tip and keeper is temporarily high, typ. ≥ 1.0 mTorr.
 - Visible arcing from the cathode tip should be observable, along with (possibly unsteady) keeper power supply voltage / current readouts.
- Ideally, the cathode tip will rapidly heat and operation will transition to constant-current mode (e.g. 1.5 A @ tens of volts).
 - If the cathode starts, the flow rate and keeper current may be reduced (e.g. to 1-3 sccm and 0.5 A) as desired soon after starting.

- **Troubleshooting**

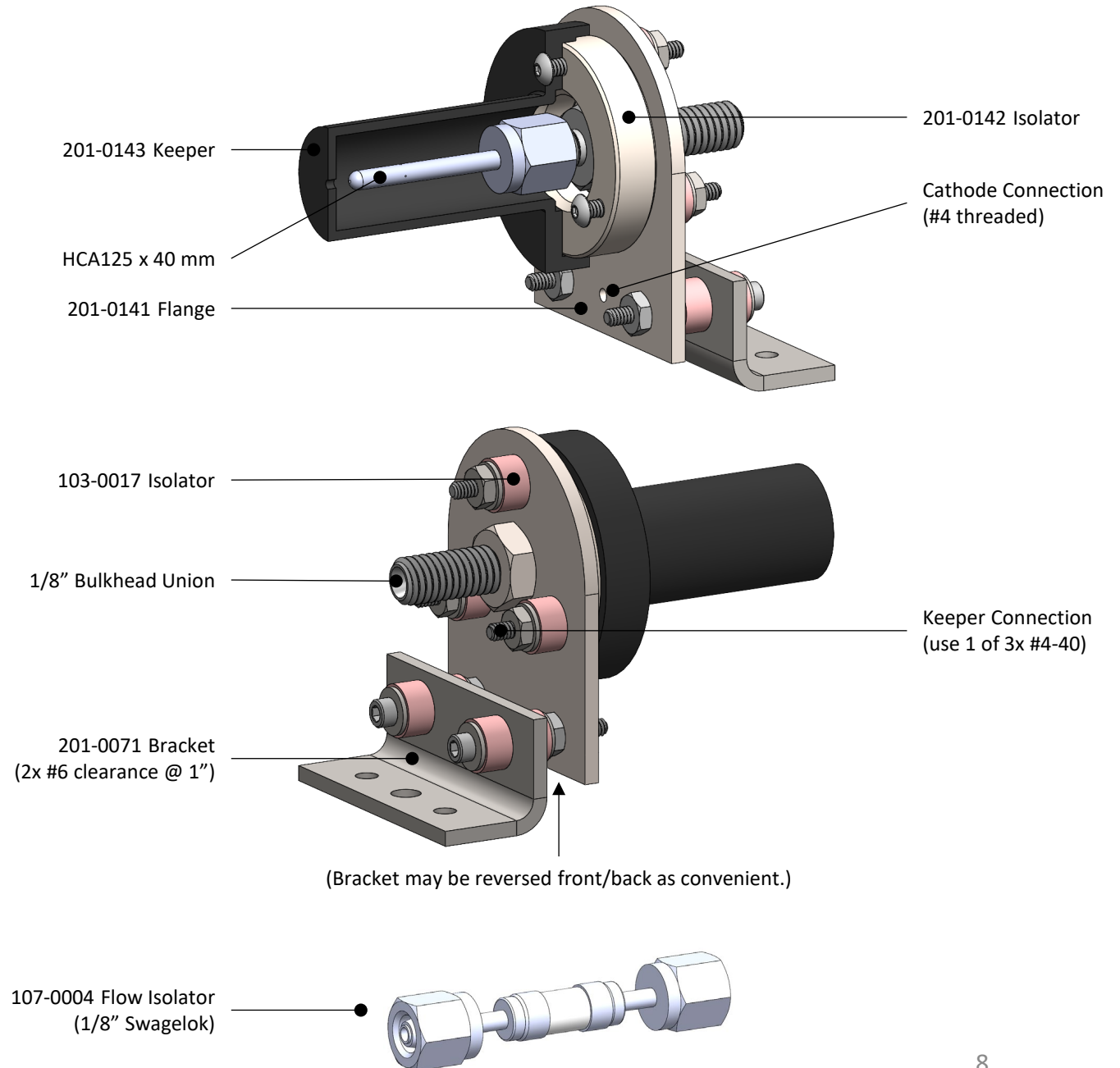
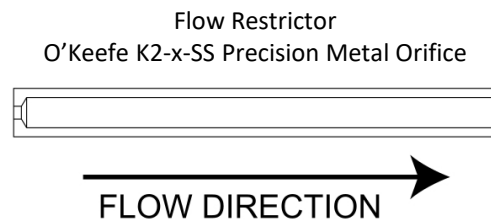
- If the cathode doesn't arc:
 - A higher peak flow is likely needed given the applied keeper voltage.
 - The peak/spike pressure on a chamber pressure gauge is usually about 5-6 mTorr before exponentially dropping to steady state.
 - The optional inline flow restrictor mainly helps to keep the pressure between the cathode tip and keeper elevated for ~ 10 s while arc heating.
 - A small restrictor (.0051") might limit the peak pressure too much, whereas a 0.007" or 0.010" restrictor will allow a higher peak pressure at the expense of a faster pressure drop.
- If the cathode arcs but doesn't stay on:
 - The starting procedure might need to be repeated a few times to soft heat the cathode and/or remove contaminants, particularly after venting.
 - It helps to have a higher current setpoint (i.e. 1.0 A to 2.0 A instead of 0.75 A or lower).
 - The first start after atmospheric exposure typically requires 1.5-2.0 A of emission current.
 - Subsequent ignitions can start at lower currents 0.5~1.0 A.
 - It may also help to have an external discharge bias supply applied if there is one available in the setup.
- Literature
 - Ryan Ham, John Williams, Seth J. Thompson and Shawn Farnell. "A Low Erosion Instant Start Ignition Process for Heaterless Hollow Cathodes," AIAA 2021-3377. AIAA Propulsion and Energy 2021 Forum. August 2021.
 - <https://arc.aiaa.org/doi/abs/10.2514/6.2021-3377>

Ignition and Operation

- Purpose: Rapid or “instant start” ignition of heaterless hollow cathode electron sources for plasma devices such as neutralizers, and ion and Hall thrusters
 - Intense plasma is formed by a combination of elevated voltage and gas flow to create a breakdown to facilitate rapid heating to thermionic emission temperatures
- References:
 - Ham, R. K., Williams, J. D., Hall, S. J., Benavides, G. F., and Verhey, T. R., “Characterization of Propellant Flow and Bias Required to Initiate an Arc Discharge in a Heaterless Hollow Cathode,” *55th Joint Propulsion Conference*, Indianapolis, IN: American Institute of Aeronautics and Astronautics, 2019.
 - Ham, R. K., Williams, J. D., Hall, S. J., Benavides, G. F., and Verhey, T. R., “Characterization of a Fixed-Volume Release System for Initiating an Arc Discharge in a Heaterless Hollow Cathode,” *IEPC-2019-890 36th International Electric Propulsion Conference*, Vienna, Austria
 - Belt, D., and J. Mankowski. 2008. "Diagnostics of the Start-Up Process of an Arc Hollow Cathode." *IEEE TRANSACTIONS ON PLASMA SCIENCE* 36 (4): 1210-1211.
 - Daykin-Iliopoulos, A., S. Gabriel, I. Golosnoy, K. Kubota, and I. Funaki. 2015. "Investigation of Heaterless Hollow Cathode Breakdown." *34th International Electric Propulsion Conference IEPC-2015-193 / ISTS-2015-b-193*.
 - Lev, D., and L. Appel. 2016 May 2-6. "Heaterless Hollow Cathode Technology - A Critical Review." *SPACE PROPULSION 2016*. MARRIOTT PARK HOTEL, ROME, ITALY.
 - Lev, D., G. Alon, D. Mikitchuk, and L. Appel. 2015. "Development of a Low Current Heaterless Hollow Cathode for Hall Thrusters." *34th International Electric Propulsion Conference IEPC-2015-163 / ISTS-2015-b-163*

Parts List

Part Number	Quantity	Material	Notes
201-0140 Assembly	1	(Assembly)	
HCA125 Hollow Cathode Tip	1	Tantalum	40 mm (1.575") length
201-0141 Flange	1	SS 304	Flat version
201-0142 Isolator	1	Alumina	
201-0143 Keeper	1	Graphite	
103-0017 Isolator	7	Alumina	#4
1/8" Bulkhead Union	1	SS 316	5182K723
201-0071 Bracket	1	SS 304	
107-0004 Flow Isolator	1	SS 304 + Alumina	1/8" Swagelok MPF A1886-2-W
Flow Restrictor	1	SS 316	O'Keefe K2-5-SS 0.0051" K2-7-SS 0.0071" K2-10-SS 0.0102"
Standard machine screws, washers, hex nuts	Multiple	SS 18-8	



Maintenance

- **Electrical Isolators**

- The **cathode** and **keeper** electrodes should be isolated from each other and **ground**, checked using an ohmmeter (i.e. ≥ 10 M Ω):
 - Ground-to-cathode
 - Ground-to-keeper
 - Cathode-to-keeper
- The electrical isolator(s) should be replaced if any partial or full shorts are detected.
- The main disc isolator (**PN 201-0142**) may be lightly grit blasted (at low pressure) if needed.

- **Propellant/Gas Feed System**

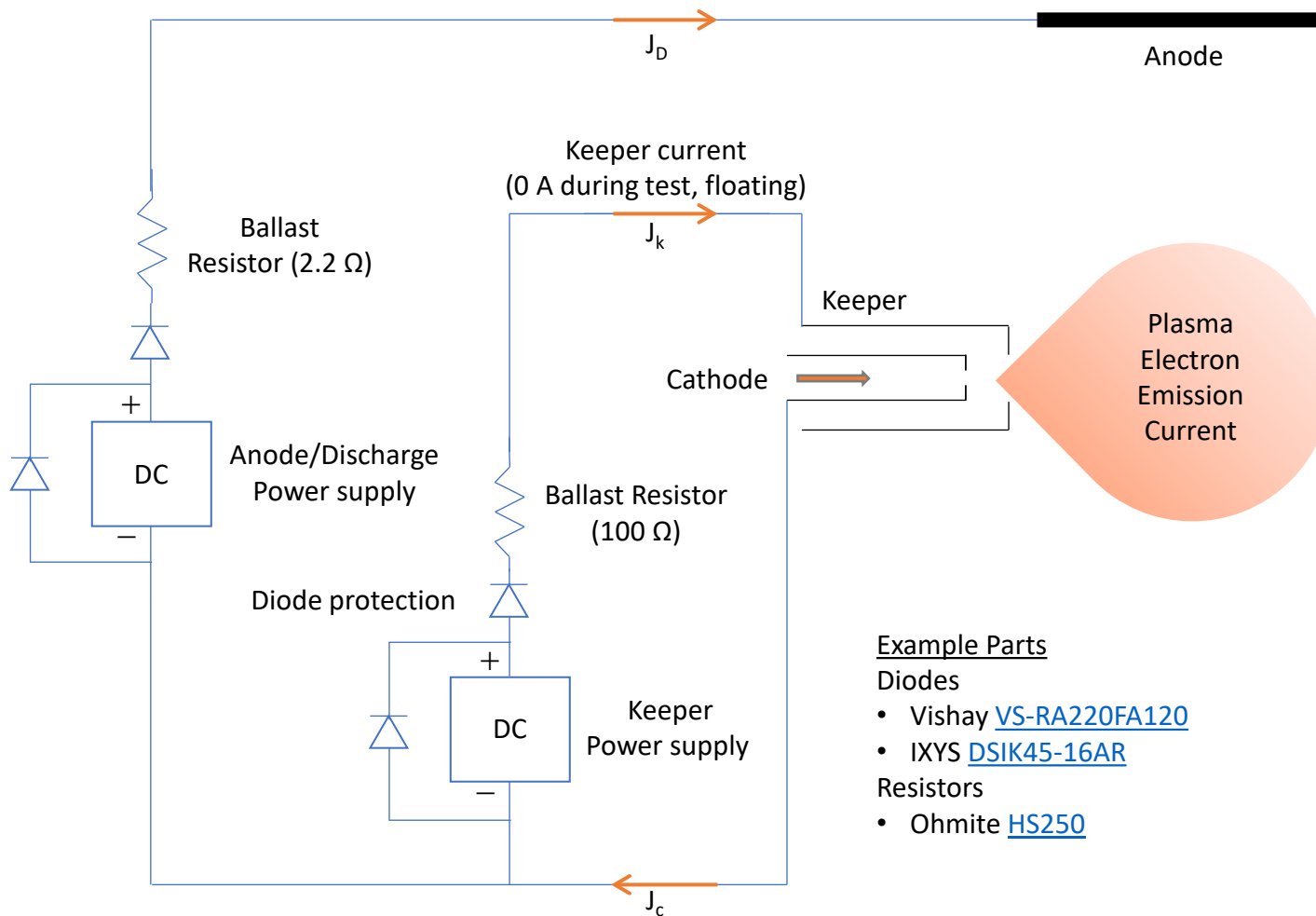
- Air leak(s) in supply lines are difficult to detect.
- Oxygen can significantly shorten cathode lifetime and/or prevent ignition.
- This is a very common troubleshooting issue.

- **Cathode Tip**

- The cathode tip may fail to ignite or become damaged from
 - Excess oxygen in the flow lines, which causes oxidation & fracturing.
 - Oxygen in the chamber environment may also shorten lifetime.
 - Sputter erosion from unusual plasma conditions.
 - Low work function material depletion at high operating temperature.
- Replacement Procedure
 - Remove the **keeper** (201-0143) and **keeper isolator** (201-0142) by unscrewing the 3x 4-40s.
 - Unscrew the old cathode tube from the bulkhead fitting.
 - Install a new tip to the bulkhead fitting.
 - Note: Only install the 1/8" Swagelok **finger tight** to ease later removal.
 - Reinstall the **keeper isolator** and **keeper**.
 - The cathode tip to keeper gap may be checked / noted using a gauge pin.
 - The keeper plate is 0.060" thick, and the target keeper-to-cathode gap is 0.1"; a gauge pin should enter about 0.16".
 - There is some tip-to-tip directional variation with the swaged connection.
 - Ideally, the cathode orifice will be fully within the keeper orifice. If not, the cathode tube may be bent by finger pressure. For instance, iteratively note a cardinal direction, disassemble, bend, and reassemble the keeper until satisfied.

Application Example

- Cathode, keeper, and anode all floating with respect to the vacuum chamber.



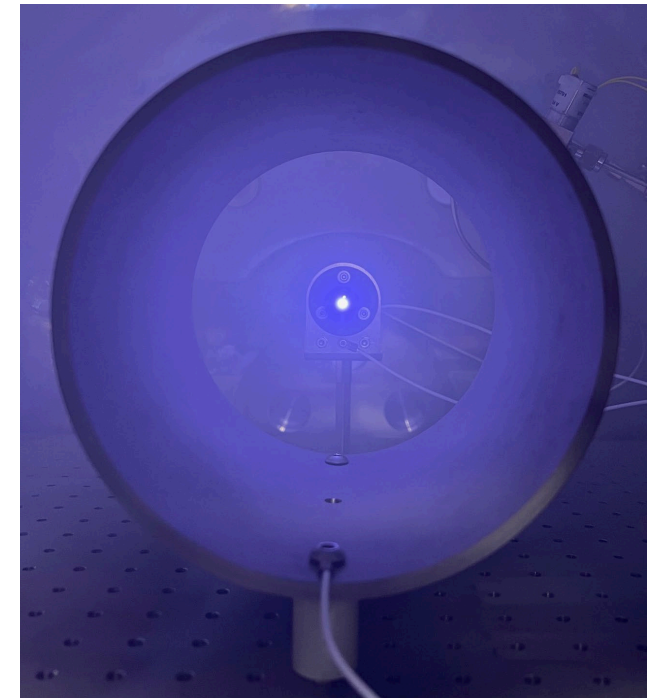
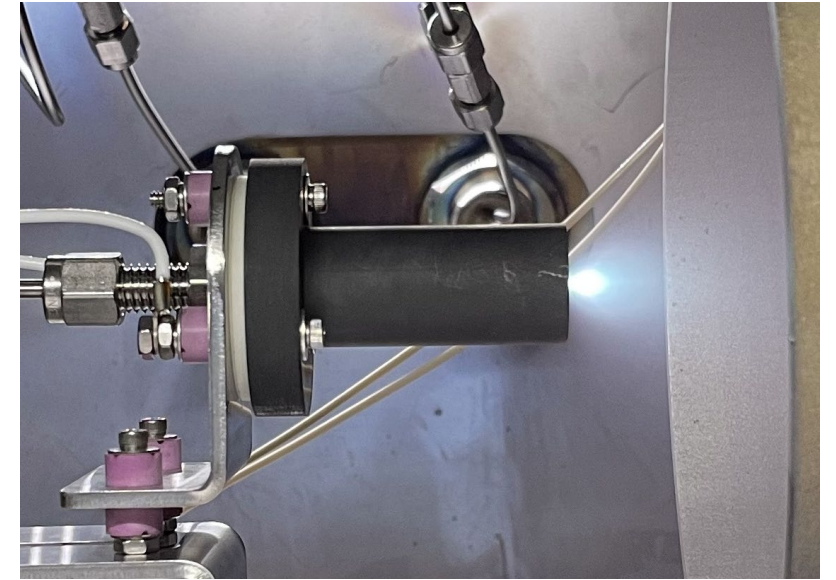
Example Parts

Diodes

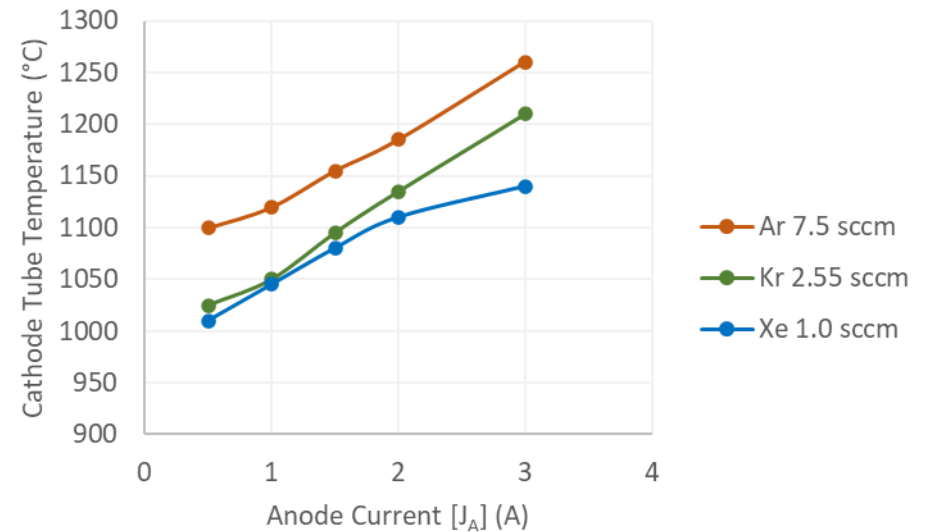
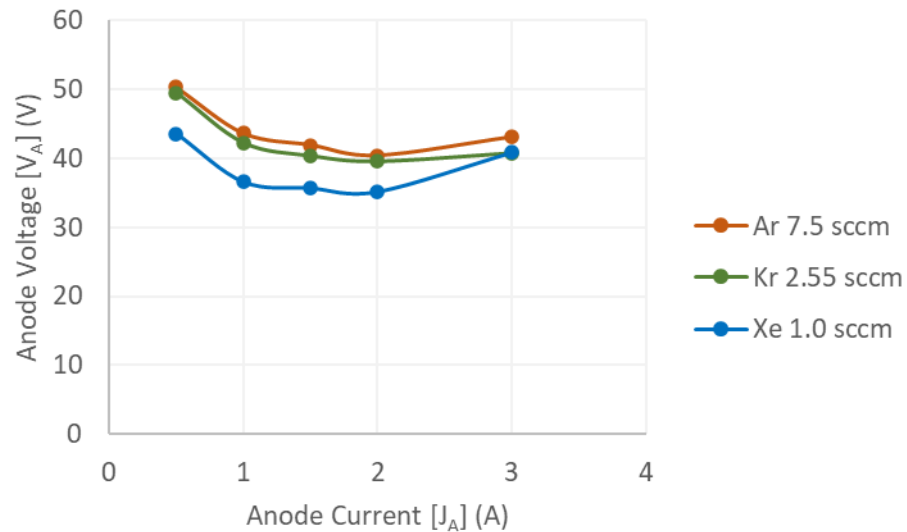
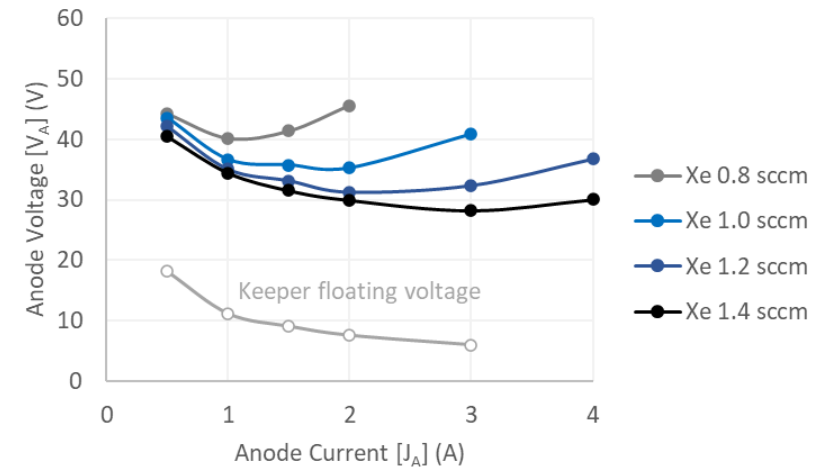
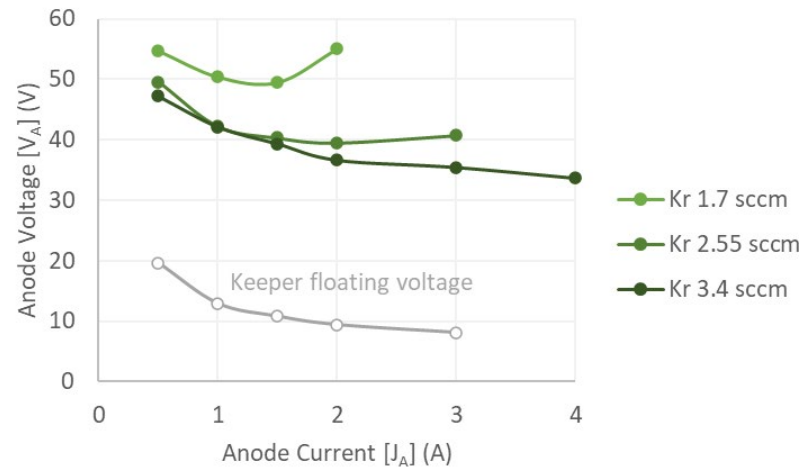
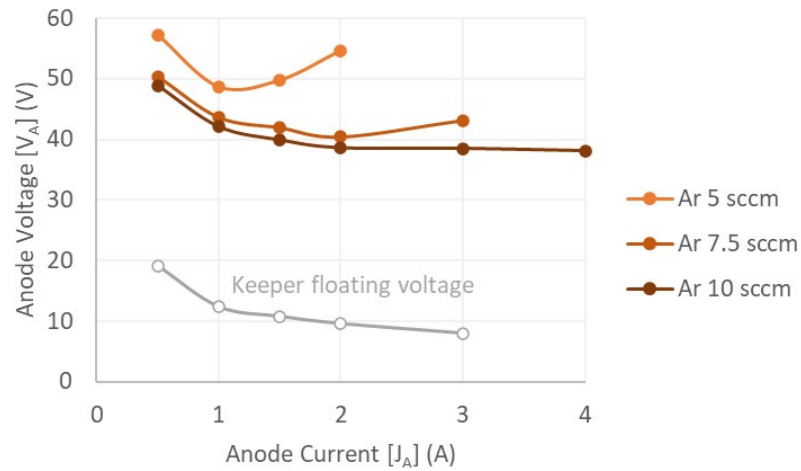
- Vishay [VS-RA220FA120](#)
- IXYS [DSIK45-16AR](#)

Resistors

- Ohmite [HS250](#)



Application Example Data



Application Example Conditions

- Cathode-to-keeper gap = 0.1"
 - Keeper orifice diameter = 0.080"
 - Keeper plate thickness = 0.060"
- Anode
 - Stainless steel cylinder, ID = 5.624" (OD = 6", wall = 0.188"), Length = 6"
 - Keeper to anode axial gap $\approx$ 2.5 cm.
- Operation at high anode voltage ($V_A > 50$ V) was avoided.
- Voltages were allowed to stabilize for 1 min. between data points.

Gas	Flow Rate [m] (sccm)	Pressure (Torr)
Argon	5	8.6×10^{-5}
	7.5	12.6
	10	16.6
Krypton	1.7	3.9
	2.55	5.8
	3.4	7.4
Xenon	0.8	2.1
	1.0	2.5
	1.2	3.0
	1.4	3.5

Revisions

- A 5/9/2024
- B 8/23/2024
 - Added “Setup” slide.
 - Added additional references on the “Ignition and Operation” slide.
- C 11/15/2024
 - Removed alternate right-angle 201-0141R Flange.
 - Changed title images for consistency.